

Q2. Merge Sort

CODE

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#include <stdio.h>

void f(int *a, int n) {
    if (n < 2) return;
    int m = n / 2, i, j, k = 0, l[m], r[n - m];
    for (i = 0; i < m; i++) l[i] = a[i];
    for (i = 0; i < n - m; i++) r[i] = a[m + i];
    f(l, m); f(r, n - m);
    for (i = 0, j = 0; k < n; k++)
        a[k] = (j == n - m || (i < m && l[i] <= r[j])) ? l[i++] : r[j++];
}

int main() {
    int a[] = {38, 27, 43, 3, 9, 82, 10}, n = 7, i;
    f(a, n);
    for (i = 0; i < n; i++) printf("%d%c", a[i], i == n - 1 ? '\n' : ' ');
    return 0;
}
```

OUTPUT 3 9 10 27 38 43 82